

KAHBOD AEINI

Computer Science Department, University of Calgary, Calgary, Alberta, Canada

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EDUCATION

University of Waterloo

Starting at Jan 2026

PhD in Electrical and Computer Engineering

Supervisor: Prof. Wojciech Golab

University of Calgary

2023 - Ongoing

Master of Computer Science

GPA: 3.825/4

Supervisor: Prof. Philipp Woelfel

Thesis: *Space-Optimal Randomized Wait-Free Lock*

Sharif University of Technology

2019 - 2023

Bachelor of Computer Engineering

GPA: 16.24/20

Thesis: *Fraud Detection from Sequences of Bank Transactions Using Various Machine Learning Algorithms*

National Organization for Development of Exceptional Talents

2013 - 2019

High School Diploma in Mathematics and Physics

GPA: 19.8/20

RESEARCH INTEREST

Distributed Computing, Concurrent & Randomized Algorithms, Data Structures, Blockchain, Database Systems, Machine Learning, Bioinformatics

PUBLICATIONS

K. Aeini, D. Bencivenga, P. Woelfel, G. Giakkoupis.

Simple and Efficient Randomized Wait-Free Lock.

Submitted to ACM-SIAM Symposium on Discrete Algorithms (SODA 2026), under review.

RESEARCH EXPERIENCE

Distributed Algorithms, Research Assistant , Graduate Student

Jun 2024 - Ongoing

Supervisor: Prof. Philipp Woelfel, University of Calgary

Designed and analyzed the first space-optimal and nearly time-optimal randomized wait-free locks in shared memory systems.

Bioinformatics, Research Assistant (Volunteer, Remote)

Feb 2023 - Aug 2023

Supervisor: Prof. Mehdi Pirooznia, Johns Hopkins University

Conducted feature selection and developed machine learning models to predict immune cell composition in lung cancer. Performed identification of immune cell types, analysis of differentially expressed genes, functional enrichment analysis, and classification of cancer subtypes using machine learning techniques.

Code Prediction, Research Assistant Machine Learning

Aug 2022- Nov 2022

Supervisor: Prof. Abbas Heydarnoori, Intelligent Software Engineering (ISE) Lab

Designed models to predict and comprehend python codes using feature extraction of Jupyter notebooks in the KGTorrent dataset.

Face Clustering, Research Assistant, Machine Learning

Oct 2021 - Feb 2022

Supervisor: Dr. Fatemeh Baharifard & Dr. Vahid Motaghed, IPM

Developed and evaluated face clustering algorithms using graph convolutional networks (GCN) for linkage prediction and pairwise classification approaches.

TEACHING EXPERIENCE

University of Calgary

2023–2025

Led and managed TA teams across 10+ undergraduate and graduate courses (up to 430 students), covering algorithms, programming, and theoretical computer science.

- **Programming Paradigms (Fall 2025)** — Managed 6 TAs, organized tutorials in Haskell & Prolog for 300+ students.
- **Randomized Algorithms (Graduate, Fall 2024)** — Delivered tutorials on randomized data structures, lower bound techniques, and complexity classes.
- **Theoretical Foundations of Computer Science II (Winter 2024 & 2025)** — Coordinated 7 TAs, supported 400+ students in automata theory and probability.
- **Python Programming (Fall 2023)** — Taught fundamentals of Python, data structures, and algorithms for 280+ students.

Sharif University of Technology

2021–2022

Served as Head TA in multiple courses, leading and mentoring 10–25 TA teams.

- **Data Structures and Algorithms** — Designed assignments & projects, led TA group of 25+.
- **Numerical Calculations** — Developed final project, led TA group of 10+.
- **Mobile Programming** — Designed and graded the final Android project.

WORK EXPERIENCE

Data Engineering, AI Lida

Feb 2022– Sep 2022

- Developed, maintained and optimized data pipelines for ETL processes using Python and PostgreSQL.
- Collaborated with the security department to gather requirements and deliver data solutions that supported business objectives.
- Interviewed, hired, trained and supervised a team of 5 data engineer interns, providing mentorship and performance feedback.

AWARDS AND SCHOLARSHIPS

Departmental Research Award, Department of Computer Science, University of Calgary

2023

Won the RA award, which is granted to the best students in each department to step away from TA responsibilities and focus on their research

Top 100 Students in Country

2019

Ranked 94 (31st in the Region) in National Universities Entrance Exam among around 160,000 participants

National Elites Fellowship

2019

Fellowship in Iran's National Elites Foundation

SKILLS

Languages: Python, R, Haskell, Java, C/C++, Prolog, Android

Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, Django, Numpy, Pandas, OpenCV, Git, LaTeX, Wordpress

Databases: PostgreSQL, MongoDB, MySQL

Other: Linux, NixOS, Kali, limma